

PROJECT TRICORDER Mk I

THEORY OF OPERATIONS

As-Built Analysis — Software 0.6.108

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This document describes how Project Tricorder Mk I actually operates as implemented in the 0.6.108 source archive. It is deliberately an as-built Theory of Operations rather than an aspirational product specification. Where the architecture contains both a reference core and device-specific implementations, the distinction is made explicitly.

1. Executive Summary

Project Tricorder Mk I is a multimodal personal field instrument implemented as an Android application whose user interface is a local WebView, whose device functions are exposed through a Kotlin JavaScript bridge, whose machine-learning engines are split between Android in-process inference and long-lived local Termux services, and whose optional context and communications layers reach outside the device only when explicitly enabled or required.

The central operating principle is not 'ask an AI a question.' It is: acquire a signal, identify the operating mode, route the signal deterministically, invoke the smallest appropriate processor, attach context, present the result, record the event, and optionally distribute a compact finding.

The system therefore behaves more like an instrument operating system than a conventional mobile application. Its principal control abstractions are Mode, Input Channel, Target Modality and Action (the tuple), an event bus, continuous sensor loops, escalation rules, operator Hold semantics, persistent field logging, and explicit communications bearers.

2. System Boundary

- The Android APK is the instrument host, operator interface, native sensor layer, storage layer, and bridge.
- The WebView UI is the primary control surface and contains the live event bus, tuple resolver, sensor-loop driver, rendering, voice command state machines, and operator controls.
- Termux is a local Linux userland used for heavyweight inference services: Gemma chat, SmolVLM vision, Stable Diffusion image generation, and Kokoro speech synthesis.
- Android in-process inference handles YAMNet audio classification and the downloaded BirdNET and iNaturalist/Coral flora-fauna models.
- Context services provide optional Earth/space/civic/flight information and are not part of the local inference core.
- Nearby Connections provides short-range tricorder-to-tricorder networking; Meshtastic over BLE provides an optional LoRa bearer for compact SUBSPACE alerts.
- The field log is the instrument's evidence record. The chat transcript is not the authoritative history.

3. Architectural Layers

3.1 Operator and Presentation Layer

The face is a single instrument-oriented phosphor UI. It exposes status lamps, mode indicators, controls, diagnostic text, sensor telemetry, cards, readings charts, logs, and voice feedback. The visual design is intentionally persistent and stateful: the display communicates whether the instrument is idle, listening, processing, holding, watching, speaking, linked, or degraded.

3.2 Deterministic Control Layer

The reference tricorder-core implements EventBus, TupleEngine, configuration loading, payload construction, backend routing, and sensor-loop semantics. The WebView contains a deliberately parallel implementation of the same EventBus/TupleEngine/SensorLoopDriver contract so the UI can operate directly. The 0.6.108 archive contains 93 passing core tests.

The important design property is that intent routing is deterministic. A button or recognized command becomes a tuple; configuration rules select the route. The language model is not asked to decide which instrument subsystem should execute the command.

3.3 Native Device Layer

MainActivity owns Android capabilities including camera/microphone permission, speech recognition, text-to-speech fallback, device motion, gravity, magnetic field, rotation vector, pressure, temperature, humidity, GNSS timing and position, battery/network status, fold posture observation, and lifecycle management. TricorderBridge exposes these capabilities to the WebView.

3.4 Local Inference Layer

The local inference stack is intentionally split. Lightweight classifiers run inside the APK. Larger generative models run as localhost HTTP services under Termux. The current launcher starts Gemma 3 4B Q4_K_M on port 8080, SmolVLM2-500M with its multimodal projector on 8081, Stable Diffusion 2.1 Turbo on 8188, and Kokoro on 5000.

3.5 Context and Communications Layer

ContextWatch polls selected external services. It can acquire weather alerts, overhead aircraft, civic/public-safety events and earthquake information. Other subsystems acquire TLEs, solar weather, geographic facts, aircraft dossiers, knowledge-bank material and translation packs.

Communications are separate from sensing. A detection does not automatically become a network packet. The Subspace layer classifies noteworthy events, applies category priority and a token rate limit, fragments the message when necessary, and transmits it through the selected bearer.

4. Fundamental Operating Model

The practical operating model can be expressed as:

**INPUT → DETECTION → CLASSIFICATION → CONTEXT → RESPONSE → MEMORY →
OPTIONAL DISTRIBUTION**

INPUT is speech, text, camera imagery, microphone audio, device motion, location, environmental sensors, external context, or a remote tricorder message. DETECTION determines that something exists. CLASSIFICATION determines what it most resembles. CONTEXT attaches location, time, identity, flight, civic, astronomical or knowledge information. RESPONSE is visual, spoken, logged or networked. MEMORY records the result. DISTRIBUTION is optional and policy-controlled.

5. Tuple and Event Operation

5.1 Tuple

The tuple is Mode + InputChannel + TargetModality + Action. The three primary configured controls are Inquiry = CHATTING/AUDIO_IN/TEXT/TOGGLE, Sensor = SCANNING/CAMERA/VISION/LOOP, and Sentry = GUARDING/CAMERA/MOTION/LOOP.

Keyword hints can change the target modality without changing the surrounding control structure. For example, image-generation language can redirect an Inquiry tuple toward the image-generation route, with chat as a configured fallback.

5.2 Event Bus

The event bus is the decoupling spine. Producers emit events such as button.press, tuple.resolved, sensor.frame, sensor.description, sensor.motion_reading, alert.raised, model.response.done and audio output events. Consumers subscribe rather than forming a dense direct-call graph.

This matters operationally because the sensing system, UI, logging and diagnostics can observe the same activity without becoming one monolithic execution path.

6. Inquiry Operations

Inquiry begins with speech or typed text. In the APK, the primary speech-input path is Android SpeechRecognizer exposed through the native bridge; the source archive also contains an STT route in the reference configuration, but the active device path is the Android recognizer rather than a separate localhost 5001 service.

- The utterance is captured and normalized.
- Special operator commands can be intercepted before generative inference.
- The text is passed to the deterministic tuple resolver.
- The selected route normally reaches the local Gemma chat service on 127.0.0.1:8080.
- Image-generation keywords can select the local Stable Diffusion route on :8188.
- The result is displayed on the instrument face.
- Speech output is produced through the Kokoro path when available, with Android TTS fallback behavior.
- The audio-sensing subsystem is held while speech output is active so the instrument does not classify its own speaker output as environmental sound.

7. Sensor Operations

Sensor mode is a continuous camera observation loop. The CameraHook obtains a rear-camera stream, captures a JPEG frame for vision, and emits sensor events around each cycle. The vision route is the fast local SmolVLM service.

The loop driver prevents overlapping executions. A new tick is not started while the previous tick for that loop is still busy. This small rule is operationally important on a phone because it prevents inference backlog from becoming an uncontrolled queue.

Sensor operation has an explicit Hold state. Holding stops the loop, freezes the operational view, and permits a still-frame analysis path. Flora/fauna classifiers can then operate on the actual image rather than on the prose generated by the vision model.

8. Sentry Operations

Sentry uses the same loop machinery but adds a motion-detection stage. Camera motion is computed by downsampling to grayscale and comparing successive frames. Device motion is independently available from the accelerometer-derived jerk signal. The configured Sentry rule can therefore combine camera and device motion.

In the current 0.6.108 UI, Sentry is deliberately configured for continuous vision behavior rather than a purely silent trigger-only guard. The architecture still contains the escalation threshold and cooldown mechanism: motion can trigger an escalation, while cooldown prevents repeated alert storms.

This is a key distinction: Sentry is not simply 'camera always on.' It is a policy-driven loop with a cheap local observation stage and a more expensive vision stage.

9. Audio Sense Operations

Audio Sense is an in-process TensorFlow Lite YAMNet pipeline. AudioRecord acquires microphone samples; YAMNet classifies short windows; the subsystem operates on a duty cycle rather than continuously. Dwell and interval are operator-adjustable.

- YAMNet provides broad environmental sound classification.
- Bird-like detections can be escalated to BirdNET for species-level classification.
- The result is written to the audio/field record and can become a Subspace candidate.
- Audio Sense can be suspended while the tricorder is speaking.
- The duty-cycle design controls battery, CPU and false self-detection pressure.

The architecture therefore treats environmental audio as a sensing instrument, not as a continuous voice conversation channel.

10. Vision and Biological Classification

The system deliberately separates general vision from specialized classification. SmolVLM produces a general description of a frame. The flora/fauna path runs dedicated TFLite classifiers against the actual bitmap. BirdNET is similarly an escalation path rather than the default classifier for every audio sample.

This separation prevents one model from being treated as a universal authority. General vision answers what appears to be in the frame; specialized classifiers supply a narrower hypothesis with a model-specific confidence score.

11. Context Acquisition

ContextWatch is a scheduled polling subsystem. It remembers the most recent usable location and uses that fix to parameterize location-dependent queries. Weather, aircraft and civic/earthquake sources have independent schedules and failure isolation.

- NOAA/NWS active alerts can be polled.
- OpenSky can be queried for nearby aircraft, with rate-limit/backoff handling.
- Civic incident and public-safety feeds can be monitored where configured.
- USGS earthquake data can be queried around the observer.
- Context events can be spoken, logged, displayed and offered to the communications layer.

External context is therefore additive. Loss of Earth link does not erase the local instrument; it removes only the context that depends on that link.

12. Knowledge and Memory

The instrument has multiple memory forms rather than one undifferentiated database.

- FieldLog is the primary append-oriented evidence record and can retain text, timestamps, GPS and media references.
- AudioLog records audio classifications and related observations.
- WatchBook retains persistent flight/civic watch material.
- DictationStore provides a separate editable spoken notebook with summaries and mind-map operations.
- KnowledgeStore provides local/remote knowledge-bank access for reference inquiries.
- BackupStore provides explicit backup/restore of instrument state.
- Readings charts aggregate field-log events into time-window and category views.

The result is closer to a field notebook plus instrument telemetry than to a disposable chat history. A later query can therefore be answered from recorded observations rather than requiring the original sensor condition to still exist.

13. Voice Output and Hands-Free Operation

Speech output is treated as an operational resource. Kokoro is the preferred local neural voice path. Android TextToSpeech provides a robust fallback/alert path. Output is queued rather than repeatedly flushed, allowing multiple alerts to coexist without one event erasing another.

Voice activity gates Audio Sense. This is a simple but important feedback-control mechanism: the instrument temporarily stops listening to a channel while it is using that channel to speak.

14. Comlink and Distributed Tricorders

14.1 Nearby Comlink

Nearby Connections provides a local peer-to-peer path. Tricorders advertise an identity containing a callsign and GUID, discover peers, establish links, exchange hello messages and can exchange JSON payloads. Peer state is tracked as seen, reachable or linked.

GPS fixes can also be exchanged, allowing one tricorder to know the reported position of another.

14.2 SUBSPACE / LoRa

The Meshtastic path is deliberately different. It is not a replacement for the richer Nearby link. Subspace is a constrained alert transport. Detections are categorized, prioritized, rate-limited with a token bucket, and fragmented into compact TF frames when necessary.

The current priority ordering places weapon, civic and flight events ahead of less urgent categories. This is a communications theory-of-operations decision: the instrument does not attempt to transmit everything it senses.

15. Operator Control, Hold and Resource Governance

The most important controls in the system are not cosmetic. They are resource governors.

- Hold stops active sensing loops and prevents runaway inference.
- Sensor and Sentry dwell controls determine observation frequency.
- Audio duty-cycle controls determine microphone/classifier utilization.
- Backend status lamps reveal whether local inference services are reachable or busy.
- Voice gating prevents self-hearing and audio contention.
- Alert cooldown prevents repetitive Sentry escalation.
- SUBSPACE rate limiting prevents a busy sensor from becoming a radio flood.
- Pinned cards prevent background sensing from erasing an operator's current object of interest.

16. Failure and Degraded Operation

The system is designed around graceful degradation rather than a single all-or-nothing service.

- If Termux is absent, the APK and local sensor/classifier functions can still exist, while chat/vision/Imagine/Kokoro dependent functions are unavailable.
- If Earth link is absent, already-downloaded local models and stores remain usable; first-run or refreshed context cannot be fetched.
- If a context provider fails, the scheduled watch continues and the failure is isolated.
- If a vision inference fails, the current frame can remain available for local handling/logging.
- If neural speech fails, Android TTS can provide a fallback path.
- If a peer disappears, the peer remains represented as seen rather than being treated as permanently linked.
- If memory pressure occurs, WebView cache and media resources can be released and neural audio playback can be stopped.

17. Local-First Does Not Mean Network-Free

The correct architectural statement is 'local-first, network-optional for many functions,' not 'the tricorder never uses a network.' Localhost model servers, Android APIs and in-process TFLite inference form the operational core. External networks are used for model/data acquisition, context services, knowledge retrieval, and optional communications.

After required packs and models are present, substantial portions of the instrument continue to operate without an Internet connection. This is materially different from a cloud-first assistant, but it is not equivalent to a completely isolated embedded appliance.

18. What Makes the Architecture a Tricorder Rather Than a Collection of Apps

The defining property is the shared operational substrate. Camera observations, audio detections, aircraft, civic alerts, astronomical data, dictation, translation and peer messages all converge on the same operator face, time base, field log and response mechanisms.

The system also distinguishes acquisition from interpretation. A camera frame can be retained even if the vision model fails. A field-log record can exist without a network. A remote alert can be received without the originating sensor being available locally. This separation is what permits the device to behave as an instrument instead of a sequence of app screens.

19. As-Built State Model

IDLE → LISTEN / SENSE / GUARD → ACQUIRE → CLASSIFY → CONTEXTUALIZE → PRESENT → SPEAK / LOG → OPTIONALLY DISTRIBUTE → RETURN TO WATCH

At any stage, a failure can terminate only the affected operation and return the instrument to a known state. Hold is a deliberate transition out of active acquisition. Termination, abort and lifecycle events similarly exist to prevent orphaned recognizers, timers, audio players or camera tracks.

20. Engineering Significance

The most significant engineering achievement in 0.6.108 is not the presence of any individual model. Each major model or data source is independently available elsewhere. The difficult part is the control system surrounding them: deterministic routing, stateful loops, rate control, specialized escalation, operator Hold, persistent evidence, local/remote separation, peer identity, and graceful failure.

In particular, the architecture demonstrates that an effective multimodal instrument does not require one enormous multimodal model. It can be constructed as a federation of small and medium specialists under a deterministic control plane.

21. Important Boundaries and Non-Claims

- The artificial horizon, heading, accelerometer, barometer and derived attitude displays are consumer-sensor visualizations, not certified navigation instruments.
- Ground speed is not indicated airspeed.
- TLE/SGP4 displays are prediction products, not radar or spacecraft telemetry.
- YAMNet/BirdNET/vision/flora classifiers provide hypotheses and confidence, not expert certification.
- External context feeds can be stale, rate-limited, unavailable or geographically incomplete.
- The system does not establish that a detected object or event is physically present with certainty merely because a model produced a label.
- The instrument is a laboratory/field appliance, not an FAA, NASA, EASA, ICAO or military-certified system.

22. Source Audit — 0.6.108

This Theory of Operations was derived from direct inspection of the 0.6.108 source archive, including the Android/Kotlin implementation, the WebView UI shell, tricorder-core, Termux launcher/bootstrap material, and the bundled operational documentation.

- MainActivity.kt — Android lifecycle, sensors, speech, TTS, networking, context integration and orchestration.
- TricorderBridge.kt — native capability boundary exposed to the WebView.
- ContextWatch.kt — scheduled external context acquisition.
- AudioSense.kt — in-process YAMNet sensing and duty cycle.
- BirdNetStore.kt / FloraStore.kt — specialist biological classification.
- FieldLog.kt / AudioLog.kt / WatchBook.kt / DictationStore.kt — persistent memory paths.
- NearbyLink.kt — peer-to-peer tricorder communications.
- MeshLink.kt / MeshProto.kt / Subspace.kt — Meshtastic/LoRa transport and rate-limited SUBSPACE.
- tricorder-core/eventBus.mjs — event spine.
- tricorder-core/tupleEngine.mjs — deterministic tuple routing.

- tricorder-core/sensorLoopDriver.mjs — scan/guard loop semantics.
- ui-shell/tricorder-ui-shell.html — active WebView control plane and instrument face.
- termux/tricorder — local inference service launcher.
- The core test suite reports 93 tests passing, 0 failures.

23. Conclusion

Project Tricorder Mk I 0.6.108 operates as a layered, deterministic, multimodal field instrument. Its sensors generate observations; specialized processors classify them; context providers enrich them; the operator interface presents them; voice and visual channels report them; persistent stores remember them; and optional Comlink/SUBSPACE bearers distribute selected findings.

The central theory is therefore simple:

THE TRICORDER IS THE ORCHESTRATION.

The models are components. The sensors are components. The feeds are components. The phone is the platform. The Tricorder emerges from the rules that cause those components to behave as one coherent instrument.

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